



Sample Application for Research Training and Career Development Funding

Through the K99/R00 Pathway to Independence Awards, NIA supports exceptional postdoctoral researchers in completing the final years of their postdoctoral work and transitioning to a role as an independent scientist. Each award has two phases, the K99 phase supporting postdoctoral training, and the R00 phase supporting an independent research career.

Copyright Notice: The awardee allows you to use the material (e.g. data, writing, graphics) in their application only for nonprofit educational purposes, provided the material remains unchanged and the principal investigator, awardee organization, and NIH NIA are credited.

**Find more NIA sample applications and information about training and career
development funding:**

<https://www.nia.nih.gov/research/training/k99-r00-sample-applications>

PI: Luth, Elizabeth	Title: Enhancing Dementia Instruction and Tools in Home Hospice Care (EDITH-HC)	
Received: 02/12/2019	FOA: PA19-129 Clinical Trial:Required	Council: 10/2019
Competition ID: FORMS-E	FOA Title: NIH Pathway to Independence Award (Parent K99/R00 - Independent Clinical Trial Required)	
1 K99 AG065624-01	Dual:	Accession Number: 4268777
IPF: 1514803	Organization: WEILL MEDICAL COLL OF CORNELL UNIV	
Former Number:	Department: Medicine-Geriatrics and Pallia	
IRG/SRG: NIA-S	AIDS: N	Expedited: N
<u>Subtotal Direct Costs</u> <u>(excludes consortium F&A)</u> Year 1: Year 2: Year 3: Year 4: Year 5: 	Animals: N Humans: Y Clinical Trial: Y Current HS Code: 30 HESC: N	New Investigator: Early Stage Investigator:
<i>Senior/Key Personnel:</i>	<i>Organization:</i>	<i>Role Category:</i>
Elizabeth Luth	Joan & Sanford I.Weill Medical College of Cornell University	PD/PI
Kathryn Bowles	Visiting Nurse Service of New York	Other (Specify)-Co-Mentor
Abraham Brody	New York University	Other (Specify)-Co-Mentor
Holly Prigerson	Joan & Sanford I.Weill Medical College of Cornell University	Other (Specify)-Mentor
Sara Czaja	Joan & Sanford I.Weill Medical College of Cornell University	Other (Specify)-Co-Mentor

Appendices

21b,21a

Always follow your funding opportunity's instructions for application format. Although these applications demonstrate good grantsmanship, time has passed since these grantees applied. The samples may not reflect the latest format or rules.

Copyright notice. The awardee allows you to use the material (e.g. data, writing, graphics) it shared in this application only for nonprofit educational purposes provided the material remains unchanged and the principal investigator, awardee organization, and NIH NIA are credited.

424 R&R and PHS-398 Specific Table Of Contents

SF 424 R&R Cover Page.....	1
Table of Contents.....	3
Performance Sites.....	4
Research & Related Other Project Information.....	5
Project Summary/Abstract(Description).....	6
Project Narrative.....	7
Bibliography & References Cited.....	8
Facilities & Other Resources.....	13
Equipment.....	17
Research & Related Senior/Key Person.....	18
Research & Related Budget Year - 1.....	56
Research & Related Budget Year - 2.....	59
Research & Related Budget Year - 3.....	62
Research & Related Budget Year - 4.....	65
Research & Related Budget Year - 5.....	68
Budget Justification.....	71
Research & Related Cumulative Budget.....	73
PHS398 Cover Page Supplement.....	74
PHS 398 Career Development Award.....	76
Candidate Information and Goals for Career Development.....	78
Specific Aims.....	84
Research Strategy.....	85
Training in the Responsible Conduct of Research.....	91
Plans and Statements of Mentor and Co-Mentor(s).....	92
Letters of Support from Collaborators,Contributors, and Consultants.....	98
Description of Institutional Environment.....	104
Institutional Commitment to Candidate's Research Career Development.....	105
PHS Human Subjects and Clinical Trials Information.....	106
Study 1: Study 1. Identifying Dementia-Related Caregiving Challenges and Strategies	108
Developing Home Hospice Training and Tool.....	108
Inclusion Enrollment Reports.....	114
Study 2: Study 2. Feasibility Testing and Finalizing a Dementia-Enhanced Training an	123
Tool for Home Hospice Clinicians.....	123
Inclusion Enrollment Reports.....	128
Study 3: Study 3. Pilot Testing Preliminary Efficacy of a Dementia-Enhanced Training	137
Tool for Home Hospice Clinicians.....	137
Inclusion Enrollment Reports.....	142
Resource Sharing.....	155
Authentication of Key Biological and/or Chemical Resources.....	156

PROJECT SUMMARY/ABSTRACT

The long-term objective of this K99/R00 application is to develop Dr. Elizabeth Luth's capacity to successfully conduct studies aimed at reducing caregiver burden and improving care for patients with Alzheimer's Disease and related dementias nearing the end of life. In the K99 phase, the proposed project supports Dr. Luth in four training objectives that will allow her to develop and transition to an independent investigator who develops culturally inclusive, practical, and scalable solutions to improve end-of-life care for patients with dementia. First, she will extend her knowledge in core substantive areas including hospice care, dementia caregiving, and recruitment and retention. Second, she will learn to develop, implement, and disseminate behavioral interventions with an emphasis on clinical care settings, workforce training, goal attainment scaling, and collaboration with community partners. Third, she will learn how to design and conduct clinical trials. Finally, for the fourth training objective, Dr. Luth will pursue professional development opportunities, specifically in the areas of grant writing and collaboration. The four research aims of this application will proceed as follows. Aim 1 will identify common challenges, strategies, and gaps in care for an understudied population; that is, community-dwelling patients with dementia near the end of life. This aim is achieved through interviewing and surveying African American and white family caregivers and hospice clinicians. Aim 2 uses key stakeholder (family caregivers, clinicians, experts) feedback to adapt dementia-focused training materials and to develop a goal assessment tool for home hospice clinicians to improve care outcomes. Aim 3 examines the feasibility and acceptability of the training and tool and revises them based on an iterative feedback process with family caregivers and clinicians. Aim 4 determines the preliminary efficacy of the training program and goal assessment tool to improve clinicians' knowledge of dementia-related challenges in home hospice care, reduce family caregiver burden, and reduce hospice disenrollment through a pilot randomized controlled trial (RCT). The proposed research works towards reducing disparities and achieving health equity by involving African American individuals in all stages of information gathering and intervention development and testing. The proposed project is consistent with the NIA's mission to conduct behavioral research on aging and foster the development of research scientists in aging. It is also aligned with the NIA's strategic goals of developing interventions to address Alzheimer's Disease and improve the health of older adults in diverse populations.

PROJECT NARRATIVE

The proposed project has the potential to increase our understanding of the challenges in caregiving for community-dwelling patients with Alzheimer's Disease and related dementias nearing the end of life and strategies used to address those challenges. The project also has the potential to improve home hospice care delivery for patients with dementia and their family caregivers. The study develops a practical, culturally inclusive training and tool for use by home hospice clinicians as they provide care to family caregivers of patients with dementia.

REFERENCES CITED

1. Luth EA. Future Time Perspective and End-of-Life Planning in Older Adults. *Research on Aging*. 2016;38(2):178-201. PMID: 26001814.
2. Luth EA, Prigerson HG. Associations between Race and Dementia Status and the Quality of End-of-Life Care. *Journal of Palliative Medicine*. 2018. PMID: 29620949.
3. Luth EA, Prigerson HG. Unintended Harm? Race Differences in the Relationship Between Advance Care Planning and Psychological Distress at the End of Life. *Journal of Pain and Symptom Management*. 2018;56(5):752-759. PMID: 30096438.
4. Phillips JA, Luth EA. Beliefs about Suicide Acceptability in the United States: How Do they Affect Suicide Mortality? *Journals of Gerontology Series B: Psychological and Social Sciences*. 2017;00(00):1-12. gbx153. DOI: <https://doi.org/10.1093/geronb/gbx153>.
5. Carr D, Luth EA. Well-Being at the End of Life. *Annual Review of Sociology*. In press;45.
6. Carr D, Luth EA. Advance Care Planning: Contemporary Issues and Future Directions. *Innovation in Aging*. 2017;1(1):igx012-igx012. DOI: <https://doi.org/10.1093/geroni/igx012>.
7. Carr DS, Luth EA. End-of-Life Planning and Health Care. In: George LK, Ferraro KF, eds. *Handbook of Aging and the Social Sciences, 8th Edition*. London: Elsevier; 2016:375-396.
8. Collins C, Ogle K. Patterns of Predeath Service Use by Dementia Patients with a Family Caregiver. *Journal of the American Geriatrics Society*. 1994;42(7):719-722. DOI: <https://doi.org/10.1111/j.1532-5415.1994.tb06530.x>.
9. Teno JM, Gozalo PL, Bynum JW, et al. Change in end-of-life care for medicare beneficiaries: Site of death, place of care, and health care transitions in 2000, 2005, and 2009. *JAMA*. 2013;309(5):470-477. PMID: 23385273.
10. Albrecht JS, Gruber-Baldini AL, Fromme EK, McGregor JC, Lee DSH, Furuno JP. Quality of Hospice Care for Patients with Dementia. *Journal of the American Geriatrics Society*. 2013;61(7):1060-1065. PMCID: PMC4006585.
11. Alzheimer's Association. 2017 Alzheimer's disease facts and figures. *Alzheimer's & Dementia*. 2017;13(4):325-373. DOI: <https://doi.org/10.1016/j.jalz.2017.02.001>.
12. De Vleminck A, Morrison RS, Meier DE, Aldridge MD. Hospice Care for Patients With Dementia in the United States: A Longitudinal Cohort Study. *Journal of the American Medical Directors Association*. 2018;19(7):633-638. PMID: [29153752](#).
13. Shega JW, Hougham GW, Stocking CB, Cox-Hayley D, Sachs GA. Patients Dying with Dementia: Experience at the End of Life and Impact of Hospice Care. *Journal of Pain and Symptom Management*. 2008;35(5):499-507. PMID:18261878.
14. Afzal N, Buhagiar K, Flood J, Cosgrave M. Quality of end-of-life care for dementia patients during acute hospital admission: a retrospective study in Ireland. *General Hospital Psychiatry*. 2010;32(2):141-146. PMID:20302987
15. Burgio KL, Williams BR, Dionne-Odom JN, et al. Racial Differences in Processes of Care at End of Life in VA Medical Centers: Planned Secondary Analysis of Data from the BEACON Trial. *Journal of Palliative Medicine*. 2016;19(2):157-163. PMID:26840851.
16. Schulz R, Martire LM. Family Caregiving of Persons With Dementia: Prevalence, Health Effects, and Support Strategies. *The American Journal of Geriatric Psychiatry*. 2004;12(3):240-249. PMID:15126224.
17. Legler A, Bradley EH, Carlson MDA. The Effect of Comorbidity Burden on Health Care Utilization for Patients with Cancer Using Hospice. *Journal of Palliative Medicine*. 2011;14(6):751-756. PMID:21548813.
18. Russell D, Diamond EL, Lauder B, et al. Frequency and Risk Factors for Live Discharge from Hospice. *Journal of the American Geriatrics Society*. 2017;65(8):1726-1732. PMID:28295138.
19. Givens JL, Kiely DK, Carey K, Mitchell SL. Healthcare proxies of nursing home residents with advanced dementia: decisions they confront and their satisfaction with decision-making. *Journal of the American Geriatrics Society*. 2009;57(7):1149-1155. PMID:19486200.
20. Miller SC, Lima JC, Looze J, Mitchell SL. Dying in U.S. nursing homes with advanced dementia: how does health care use differ for residents with, versus without, end-of-life Medicare skilled nursing facility care? *Journal of Palliative Medicine*. 2012;15(1):43-50. PMID:22175816.

21. Lopez RP, Mitchell SL, Givens JL. Preventing Burdensome Transitions of Nursing Home Residents with Advanced Dementia: It's More than Advance Directives. *Journal of Palliative Medicine*. 2017;20(11):1205-1209. DOI: <https://doi.org/10.1089/jpm.2017.0050>
22. Mitchell SL, Kiely DK, Jones RN, Prigerson H, Volicer L, Teno JM. Advanced dementia research in the nursing home: the CASCADE study. *Alzheimer disease and associated disorders*. 2006;20(3):166-175. PMID:16917187.
23. Powers RE, Herrington HL. Hospice Dementia Care. In: Boltz M, Galvin JE, eds. *Dementia Care: An Evidence-Based Approach*. Switzerland: Springer International Publishing; 2016:261-298. DOI: https://doi.org/10.1007/978-3-319-18377-0_16.
24. Unroe KT, Meier DE. Quality of Hospice Care for Individuals with Dementia. *Journal of the American Geriatrics Society*. 2013;61(7):1212-1214. PMID:23855849.
25. Sachs GA, Shega JW, Cox-Hayley D. Barriers to excellent end-of-life care for patients with dementia. *Journal of General Internal Medicine*. 2004;19(10):1057-1063. PMID:15482560.
26. Lawrence V, Samsi K, Murray J, Harari D, Banerjee S. Dying well with dementia: Qualitative examination of end-of-life care. *British Journal of Psychiatry*. 2018;199(5):417-422. PMID:21947653.
27. Harris-Kojetin L, Sengupta M, Park-Lee E, Valverde R, Caffrey VR, Lendon J. *Long-term care providers and services users in the United States: Data from the National Study of Long-Term Care Providers, 2013–2014*. National Center for Health Statistics;2016. PMID:27023287.
28. Onken LS, Carroll KM, Shoham V, Cuthbert BN, Riddle M. Reenvisioning Clinical Science: Unifying the Discipline to Improve the Public Health. *Clinical Psychological Science*. 2013;2(1):22-34. PMID:25821658.
29. Hebert LE, Weuve J, Scherr PA, Evans DA. Alzheimer disease in the United States (2010–2050) estimated using the 2010 census. *Neurology*. 2013;80(19):1778-1783. PMID:23390181.
30. Wright AA, Zhang B, Ray A, et al. Associations between end-of-life discussions, patient mental health, medical care near death, and caregiver bereavement adjustment. *JAMA*. 2008;300(14):1665-1673. doi:10.1001/jama.300.14.1665.
31. McCarthy M, Addington-Hall J, Altmann D. The experience of dying with dementia: A retrospective study. *International Journal of Geriatric Psychiatry*. 1998;12(3):404-409. PMID:9152728.
32. Riffin C, Van Ness PH, Wolff JL, Fried T. Family and Other Unpaid Caregivers and Older Adults with and without Dementia and Disability. *Journal of the American Geriatrics Society*. 2017;65(8):1821-1828. PMID:28426910.
33. Kasper JD, Freedman VA, Spillman BC, Wolff JL. The Disproportionate Impact Of Dementia On Family And Unpaid Caregiving To Older Adults. *Health Affairs*. 2015;34(10):1642-1649. PMID:26438739.
34. National Alliance for C. *Dementia Caregiving in the U.S*. Bethesda, MD February 2017 2017.
35. Haley WE, Gitlin LN, Wisniewski SR, et al. Well-being, appraisal, and coping in African-American and Caucasian dementia caregivers: findings from the REACH study. *Aging & Mental Health*. 2004;8(4):316-329. PMID:15370048.
36. National Academies of Sciences E, Medicine. *Families caring for an aging America*. National Academies Press; 2016. DOI: <https://doi.org/10.17226/23606>.
37. Lines LW, JM. *Racial and Ethnic Disparities in Alzheimer's Disease: A Literature Review*. February 2014. RTI Press publication No. RR-0024-1412. Research Triangle Park, NC: RTI Press. <https://www.rti.org/rti-press>. Accessed February 5, 2019.
38. Black BS, Johnston D, Rabins PV, Morrison A, Lyketsos C, Samus QM. Unmet Needs of Community-Residing Persons with Dementia and Their Informal Caregivers: Findings from the Maximizing Independence at Home Study. *Journal of the American Geriatrics Society*. 2013;61(12):2087-2095. PMID:24479141.
39. von Kutzleben M, Schmid W, Halek M, Holle B, Bartholomeyczik S. Community-dwelling persons with dementia: What do they need? What do they demand? What do they do? A systematic review on the subjective experiences of persons with dementia. *Aging & Mental Health*. 2012;16(3):378-390. PMID:22250961.
40. Miranda-Castillo C, Woods B, Orrell M. The needs of people with dementia living at home from user, caregiver and professional perspectives: a cross-sectional survey. *BMC Health Services Research*. 2013;13(1):43. PMID:23379786.

41. Moore MJ, Zhu CW, Clipp EC. Informal Costs of Dementia Care: Estimates From the National Longitudinal Caregiver Study. *The Journals of Gerontology: Series B*. 2001;56(4):S219-S228. PMID:11445614.
42. Schulz R, O'Brien AT, Bookwala J, Fleissner K. Psychiatric and Physical Morbidity Effects of Dementia Caregiving: Prevalence, Correlates, and Causes. *The Gerontologist*. 1995;35(6):771-791. PMID:8557205.
43. Dassel KB, Carr DC. Does Dementia Caregiving Accelerate Frailty? Findings From the Health and Retirement Study. *The Gerontologist*. 2016;56(3):444-450. PMID:25161263.
44. Vitaliano PP, Zhang J, Scanlan JM. Is caregiving hazardous to one's physical health? A meta-analysis. *Psychological bulletin*. 2003;129(6):946. PMID:14599289.
45. AGS Panel on Persistent Pain in Older Persons. The management of persistent pain in older persons. *Journal of the American Geriatrics Society*. 2002;50(6 Suppl):S205-224. PMID:12067390.
46. Cintron A, Morrison RS. Pain and Ethnicity in the United States: A Systematic Review. *Journal of Palliative Medicine*. 2006;9(6):1454-1473. PMID:17187552.
47. Hoffman KM, Trawalter S, Axt JR, Oliver MN. Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites. *Proceedings of the National Academy of Sciences*. 2016;113(16):4296-4301. PMID: [27044069](#).
48. Trevino KM, Zhang B, Shen MJ, Prigerson HG. Accuracy of advanced cancer patients' life expectancy estimates: The role of race and source of life expectancy information. *Cancer*. 2016;122(12):1905-1912. PMID:27028270.
49. Weeks JC, Cook E, O'Day SJ, et al. Relationship between cancer patients' predictions of prognosis and their treatment preferences. *JAMA*. 1998;279(21):1709-1714. PMID:9624023.
50. Mack JW, Weeks JC, Wright AA, Block SD, Prigerson HG. End-of-life discussions, goal attainment, and distress at the end of life: predictors and outcomes of receipt of care consistent with preferences. *Journal of Clinical Oncology*. 2010;28(7):1203-1208. PMID:20124172.
51. Shega JW, Tozer C. Improving the care of people with dementia at the end of life: The role of hospice and the US experience. *Dementia*. 2009;8(3):377-389. DOI: <https://doi.org/10.1177/1471301209104979>.
52. Kverno KS, Black BS, Blass DM, Geiger-Brown J, Rabins PV. Neuropsychiatric Symptom Patterns in Hospice-Eligible Nursing Home Residents with Advanced Dementia. *Journal of the American Medical Directors Association*. 2008;9(7):509-515. PMID:18755425.
53. Desai AK, Grossberg GT. Recognition and Management of Behavioral Disturbances in Dementia. *Primary Care Companion to The Journal of Clinical Psychiatry*. 2001;3(3):93-109. PMID:15014607.
54. Brodaty H, Arasaratnam C. Meta-Analysis of Nonpharmacological Interventions for Neuropsychiatric Symptoms of Dementia. *American Journal of Psychiatry*. 2012;169(9):946-953. PMID:22952073.
55. Voisin T, Andrieu S, Cantet C, Vellas B. Predictive factors of hospitalizations in Alzheimer's disease: a two-year prospective study in 686 patients of the REAL.FR study. *The Journal of Nutrition, Health & Aging*. 2010;14(4):288-291. PMID:20305996.
56. Becker MA, Boaz TL, Andel R, Gum AM, Papadopoulos AS. Predictors of Preventable Nursing Home Hospitalizations: The Role of Mental Disorders and Dementia. *The American Journal of Geriatric Psychiatry*. 2010;18(6):475-482. DOI: <https://doi.org/10.1097/JGP.0b013e3181b2145a>.
57. Soto ME, Andrieu S, Villars H, et al. Improving Care of Older Adults With Dementia: Description of 6299 Hospitalizations over 11 Years in a Special Acute Care Unit. *Journal of the American Medical Directors Association*. 2012;13(5):486.e481-486.e486. DOI: <https://doi.org/10.1016/j.jamda.2011.12.058>.
58. Gaugler JE, Wall MM, Kane RL, et al. The Effects of Incident and Persistent Behavioral Problems on Change in Caregiver Burden and Nursing Home Admission of Persons with Dementia. *Medical Care*. 2010;48(10):875-883. PMID:20733529.
59. Gitlin LN, Hodgson NA, Choi SSW. Home-Based Interventions Targeting Persons with Dementia: What Is the Evidence and Where Do We Go from Here? In: Boltz M, Galvin JE, eds. *Dementia Care: An Evidence-Based Approach*. Springer, Cham; 2016:167-188. DOI: https://doi.org/10.1007/978-3-319-18377-0_11.

60. Schulz R, Burgio L, Burns R, et al. Resources for Enhancing Alzheimer's Caregiver Health (REACH): Overview, Site-Specific Outcomes, and Future Directions. *The Gerontologist*. 2003;43(4):514-520. PMID:12937330.
61. Czaja SJ, Gitlin LN, Schulz R, et al. Development of the Risk Appraisal Measure: A Brief Screen to Identify Risk Areas and Guide Interventions for Dementia Caregivers. *Journal of the American Geriatrics Society*. 2009;57(6):1064-1072. PMID:19453305.
62. Czaja SJ, Lee CC, Perdomo D, et al. Community REACH: An Implementation of an Evidence-Based Caregiver Program. *The Gerontologist*. 2018;58(2):e130-e137. PMID:29562361.
63. Gitlin LN, Marx K, Stanley IH, Hodgson N. Translating Evidence-Based Dementia Caregiving Interventions into Practice: State-of-the-Science and Next Steps. *The Gerontologist*. 2015;55(2):210-226. DOI: <https://doi.org/10.1093/geront/gnu123>.
64. Mitchell SL, Kiely DK, Hamel MB. Dying with advanced dementia in the nursing home. *Archives of Internal Medicine*. 2004;164(3):321-326. PMID:14769629.
65. Mitchell SL, Morris JN, Park PS, Fries BE. Terminal care for persons with advanced dementia in the nursing home and home care settings. *Journal of Palliative Medicine*. 2004;7(6):808-816. PMID:15684848.
66. Goodman C, Evans C, Wilcock J, et al. End of life care for community dwelling older people with dementia: an integrated review. *International Journal of Geriatric Psychiatry*. 2009;25(4):329-337. PMID:19688739.
67. Rockwood K, Howlett S, Stadnyk K, Carver D, Powell C, Stolee P. Responsiveness of goal attainment scaling in a randomized controlled trial of comprehensive geriatric assessment. *Journal of Clinical Epidemiology*. 2003;56(8):736-743. DOI:[https://doi.org/10.1016/S0895-4356\(03\)00132-X](https://doi.org/10.1016/S0895-4356(03)00132-X).
68. Rockwood K, Stolee P, Howard K, Mallery L. Use of Goal Attainment Scaling to Measure Treatment Effects in an Anti-Dementia Drug Trial. *Neuroepidemiology*. 1996;15(6):330-338. PMID:8930946.
69. Rockwood K, Stolee P, FoxP RA. Use of goal attainment scaling in measuring clinically important change in the frail elderly. *Journal of Clinical Epidemiology*. 1993;46(10):1113-1118. PMID:8410096.
70. Burnes D, Connolly M-T, Hamilton R, Lachs MS. The feasibility of goal attainment scaling to measure case resolution in elder abuse and neglect adult protective services intervention. *Journal of Elder Abuse & Neglect*. 2018;30(3):209-222. PMID:29601281.
71. Burnes D, Lachs MS. The Case for Individualized Goal Attainment Scaling Measurement in Elder Abuse Interventions. *Journal of Applied Gerontology*. 2015;36(1):116-122. PMID:25873451.
72. Pillemer K, Suitor JJ, Henderson JCR, et al. A Cooperative Communication Intervention for Nursing Home Staff and Family Members of Residents. *The Gerontologist*. 2003;43(suppl_2):96-106. DOI: https://doi.org/10.1093/geront/43.suppl_2.96.
73. Robison J, Curry L, Gruman C, Porter M, Henderson JCR, Pillemer K. Partners in Caregiving in a Special Care Environment: Cooperative Communication Between Staff and Families on Dementia Units. *The Gerontologist*. 2007;47(4):504-515. DOI: <https://doi.org/10.1093/geront/47.4.504>.
74. Teresi JA, Ramirez M, Ellis J, et al. A staff intervention targeting resident-to-resident elder mistreatment (R-REM) in long-term care increased staff knowledge, recognition and reporting: Results from a cluster randomized trial. *International Journal of Nursing Studies*. 2013;50(5):644-656. PMID:23159018.
75. Bluestone J, Johnson P, Fullerton J, Carr C, Alderman J, BonTempo J. Effective in-service training design and delivery: evidence from an integrative literature review. *Human resources for health*. 2013;11:51-51. PMID:24083659.
76. Morse JM. Designing funded qualitative research. In: Denzin N, Lincoln Y, eds. *Handbook of Qualitative Research*. Thousand Oaks, CA: Sage Publications; 1994:220-235.
77. de Meyrick J. The Delphi method and health research. *Health Education*. 2003;103(1):7-16. DOI: <https://doi.org/10.1108/09654280310459112>.
78. Roberts K, Holland J, Prigerson HG, et al. Development of the Bereavement Risk Inventory and Screening Questionnaire (BRISQ): Item generation and expert panel feedback. *Palliative Supportive Care*. 2017;15(1):57-66. PMID: [27516152](https://doi.org/10.27516/152).
79. Fonteyn ME, Kuipers B, Grobe SJ. A Description of Think Aloud Method and Protocol Analysis. *Qualitative Health Research*. 1993;3(4):430-441. DOI: <https://doi.org/10.1177/104973239300300403>.

80. Czaja SJ, Charness N, Fisk AD, et al. Factors predicting the use of technology: findings from the Center for Research and Education on Aging and Technology Enhancement (CREATE). *Psychology and aging*. 2006;21(2):333-352. PMID:16768579.
81. Hertzog MA. Considerations in determining sample size for pilot studies. *Research in Nursing & Health*. 2008;31(2):180-191. PMID:18183564.
82. Connelly LM. Pilot studies. *MedSurg Nursing*. 2008;17(6):411-412. PMID:19248407.
83. Bothwell LE, Greene JA, Podolsky SH, Jones DS. Assessing the Gold Standard — Lessons from the History of RCTs. *New England Journal of Medicine*. 2016;374(22):2175-2181. PMID:27248626.
84. Annear MJ, Toyne CM, Eccleston CE, et al. Dementia Knowledge Assessment Scale: Development and Preliminary Psychometric Properties. *Journal of the American Geriatrics Society*. 2015;63(11):2375-2381. DOI: <https://doi.org/10.1111/jgs.13707>.
85. Higginson IJ, Gao W, Jackson D, Murray J, Harding R. Short-form Zarit Caregiver Burden Interviews were valid in advanced conditions. *Journal of Clinical Epidemiology*. 2010;63(5):535-542. PMID:19836205.
86. Yu J, Yap P, Liew TM. The optimal short version of the Zarit Burden Interview for dementia caregivers: diagnostic utility and externally validated cutoffs. *Aging & Mental Health*. 2018:1-5. PMID:29553806.
87. Pargament K, Feuille M, Burdzy D. The Brief RCOPE: Current Psychometric Status of a Short Measure of Religious Coping. *Religions*. 2011;2(1). DOI:[10.3390/rel2010051](https://doi.org/10.3390/rel2010051).
88. Brondolo E, Kelly KP, Coakley V, et al. The Perceived Ethnic Discrimination Questionnaire: Development and Preliminary Validation of a Community Version. *Journal of Applied Social Psychology*. 2005;35(2):335-365. DOI: <https://doi.org/10.1111/j.1559-1816.2005.tb02124.x>
89. Cunningham WA, Preacher KJ, Banaji MR. Implicit Attitude Measures: Consistency, Stability, and Convergent Validity. *Psychological Science*. 2001;12(2):163-170. PMID:11340927.
90. Project Implicit. 2011; <https://implicit.harvard.edu/implicit/index.jsp>. Accessed January 15, 2019.
91. Turner-Stokes L. Goal attainment scaling (GAS) in rehabilitation: a practical guide. *Clinical Rehabilitation*. 2009;23(4):362-370. PMID:19179355.
92. Hausmann LRM, Jeong K, Bost JE, Ibrahim SA. Perceived discrimination in health care and health status in a racially diverse sample. *Medical care*. 2008;46(9):905-914. PMID: [18725844](https://pubmed.ncbi.nlm.nih.gov/18725844/).
93. Prigerson HG. Socialization to dying: social determinants of death acknowledgement and treatment among terminally ill geriatric patients. *Journal of Health and Social Behavior*. 1992;33(4):378-395. PMID:1464721.
94. Brigham JC. College Students' Racial Attitudes. *Journal of Applied Social Psychology*. 1993;23(23):1933-1967. DOI: <https://doi.org/10.1111/j.1559-1816.1993.tb01074.x>
95. Rubin HJ, Rubin IS. *Qualitative interviewing: The art of hearing data*. Sage; 2005.
96. Kamal AH, Bull J, Wolf S, et al. Characterizing the Hospice and Palliative Care Workforce in the U.S.: Clinician Demographics and Professional Responsibilities. *Journal of Pain and Symptom Management*. 2016;51(3):597-603. PMID:26550934.
97. Workforce HH. *Nursing Workforce Projections by Ethnicity and Race 2014-2030*. Health Resources and Services Administration;2017. <https://bhw.hrsa.gov/national-center-health-workforce-analysis>. Accessed February 5, 2019.

Candidate Background

My path to independent investigator is the confluence of personal, professional, and training experiences. My interest in end-of-life (EOL) experiences, and specifically for persons with Alzheimer's Disease and related dementias, began long before I became a researcher. Between 2001 and 2014, three of my grandparents died, including my maternal grandfather, who died after a decade-long battle with Alzheimer's Disease. His wife was his only, 24-hour a day, caregiver during his illness. I was fortunate enough to spend time with each of my grandparents before they died. However, even in my grief, I realized that not all families were as lucky as mine. Had my father not been able to afford paid caregivers, would he have filled the gaps that so often fall to adult daughters? Would he have been as accepting of my grandmother's death if she had not been comforted in her final days by a visit from her beloved parish priest? What if my maternal grandmother had not been at my grandfather's side during his descent into dementia? During this time period, my concurrent professional experience in different non-profit service delivery settings instilled in me two convictions. First, to be adopted, solutions to service and care delivery problems must be practical and complement traditional care. Second, working with vulnerable, underserved populations often requires different and creative approaches.

I returned to graduate school to train as a medical sociologist to enhance my understanding of EOL processes and outcomes, particularly for older adults who make up 75% of U.S. deaths annually. Graduate school provided theoretical frameworks, vocabulary, and methodological training to explain and document the ways disparities in EOL experiences manifest themselves. My sociological training helped me to situate individuals' experiences within the multi-level and dynamic contexts of family, organization, neighborhood, and social structure that shape the choices they perceive and have available to them. I came to understand why and how sociodemographic disparities in health outcomes result from both individual choices and the constraints and dynamics of the social structures in which we are embedded. I also received rigorous methodological training in qualitative methods and advanced statistical techniques, the latter of which I have used in 3 first-authored publications on psychosocial and race differences in advance care planning and the quality of EOL care.¹⁻³

Upon completing my doctorate, I intentionally took a different path than many individuals with PhDs in sociology. Rather than pursuing a teaching position in a Sociology Department, I committed fully to a career in clinical research and applied for a T32 Postdoctoral Fellow post in an academic medical school. Since September 2017, Weill Cornell Medicine (WCM) has provided an ideal setting for applying the conceptual and methodological tools from my sociology training to rigorous research focused on caring for sick and dying patients. I have leveraged the support and resources at WCM to continue to grow as a researcher. Developing my grant writing skills, I obtained two grants from WCM and Visiting Nurse Service of New York (VNSNY) not only to continue my research on sociodemographic (e.g. race, socioeconomic status, gender) disparities in EOL care, but also to advance my skills in four specific ways. First, the studies have allowed me to extend my focus beyond EOL care quality to other aspects of the EOL: hospice and hospital care. Second, I have been able to concentrate on a specific patient population: patients with Alzheimer's Disease and related dementias (PwD). Third, I am gaining a deeper understanding of the many challenges that occur as a consequence of dementia caregiving through pilot interviews with clinicians and family caregivers (FCG) of PwD. Fourth, I am leading collaborative teams of co-Investigators across NewYork Presbyterian hospitals and with VNSNY in order to complete these studies. I have been productive during my T32 Postdoctoral Fellowship, having four articles (2 first-authored) accepted for publication.²⁻⁵

My personal and professional experiences, graduate training as a medical sociologist, and T32 postdoctoral training put me in an excellent position to approach problems as complex, multi-layered, multi-leveled, requiring practical, clinically useful solutions. My past and current research reflects my deep commitment to developing solutions that reduce sociodemographic disparities and promote equity in health outcomes. My background, training, and accomplishments to date ideally position me to develop a sustained career as an independent investigator. The proposed application will allow me to further develop knowledge and skills needed to launch as an independent investigator who develops culturally inclusive, practical, and scalable solutions to improve EOL care for PwD and FCG. First, I will benefit from experience in substantive topics related to my areas of research including hospice, dementia caregiving, and recruitment and retention. Second, although I have expertise in sociological theory and methodologies, which enhance my ability to understand and analyze complex problems, I wish to gain expertise in applying these models and methods to real-world problems. To address this gap, I require additional training in behavioral intervention development, dissemination of results, and designing and conducting randomized controlled trials (RCTs) and pragmatic trials. Finally, although I have worked to develop my grant writing skills and form collaborative networks, I seek additional training in these areas to develop successful R00 and competitive R01 applications and to build collaborations to mount multi-site, large scale trials. The K99/R00 Pathway to Independence Award is therefore an ideal vehicle to provide me with the time, mentorship, and training support I need to become an independent, R01-funded investigator.

Career Goals and Objectives

My long-term objective is to become a leader in developing culturally inclusive, clinically relevant, scalable solutions to improve EOL care and support for patients with Alzheimer's Disease and related dementias (PwD) and their family caregivers (FCG). I am committed to developing solutions that reduce sociodemographic disparities and promote equity in health outcomes. During my doctoral training in sociology I gained a theoretical understanding of the individual factors and structural processes that contribute to disparities in health outcomes. I also received training in advanced statistical techniques including multilevel modeling and latent class analysis. I applied my graduate training to complex analyses of secondary survey data to understand psychosocial and sociodemographic disparities in older adults' advance care planning completion^{1,3,6,7} and EOL care quality.^{2,3} I also received training in qualitative methods, leading an interview study. As a T32 Postdoctoral Fellow, with the support of my mentor, Dr. Prigerson, I have published two first-authored papers and am leading studies to broaden and deepen my understanding of EOL care in two ways. First, I have expanded my focus within the area of EOL care to encompass hospice and hospital care. Second, I have learned more about Geriatrics clinical care, while increasing my focus on a specific population: patients with dementia. In my postdoctoral research to date, I have applied my quantitative analysis skills to examine sociodemographic disparities in secondary survey data and hospice and hospital electronic records. I am further developing my qualitative skills in a pilot interview study of FCG of PwD. To reach my long-term objective of becoming an expert in improving care and support for dying PwD and FCG, I need additional knowledge and skills that I propose to acquire in the K99 phase and apply in the R00 phase of the project. The career development plan I outline here will provide me with needed support from primary and co-mentors during the K99 phase of the project, while also supporting my transition to research independence in the R00 phase. The additional knowledge and skills I will acquire as part of my career development plan (outlined below) will allow me to conduct the study proposed in the R00 and in future R01-level randomized controlled and pragmatic trials. In sum, this K99/R00 Pathway to Independence Award will provide me with the training, experiences, and preliminary work I need to launch as an independent investigator focused on improving EOL care for PwD.

Training Objective 1. Extend knowledge in core substantive areas: hospice, dementia caregiving, and recruitment and retention. My graduate work was anchored in analyzing secondary data to improve understanding of sociodemographic disparities in EOL care quality. My postdoctoral work has added dementia status as a key explanatory factor for differences in hospital and hospice care outcomes. During the K99 portion of this project, I will engage with mentors and advisors and participate in didactic activities such as shadowing providers and attending grand rounds and team meetings to deepen my understanding in key areas. These key areas include: the clinical and real-world aspects of hospice care and dementia caregiving, how hospice organizations operate and deliver care, and recruitment and retention of persons who are part of vulnerable, hard-to-reach, and underrepresented groups, such as hospice patients, PwD, and African American persons. Learning about hospice operations and care delivery will enable me to design interventions that complement existing practices. Increasing my understanding of dementia caregiving will allow me to develop interventions that are responsive to the unique challenges of caring for dying PwD in community settings. Expanding my knowledge and experience recruiting and working with individuals from underrepresented groups will ensure I can continue to work towards achieving health equity by incorporating their perspectives into all stages of intervention development in this and future projects.

Training Objective 2. Develop, implement, and disseminate behavioral interventions. Before earning my doctorate, my professional career working for community-based non-profit organizations was devoted to helping program staff develop and implement techniques to perform program evaluations. As a result of that experience, I have a deep appreciation that any steps adding to routine service delivery must be practical and feasible for the particular service setting. However, I am not familiar with best practices for developing rigorous and robust interventions, particularly in the areas of clinical care settings (such as hospice), workforce training, goal attainment scaling, and collaborating with community partners. During the K99 portion of this project I will participate in formal didactic experiences including coursework and training seminars to learn how to develop and implement behavioral interventions for older adults. I will also engage with mentors and advisors who are experts in this area. I will use this knowledge during the K99 phase to develop a training and tool-based intervention for home hospice clinicians to support FCG of PwD.

Training Objective 3. Design and conduct clinical trials: RCT and pragmatic trials. My graduate training provided me with expertise in designing and conducting research studies involving qualitative data collection (interviews, focus groups) and collection, management, and analysis of quantitative data. The K99 portion of this project will provide me with the opportunity to develop expertise in designing and conducting RCT and pragmatic trials, which ultimately will allow me to pilot test and rigorously measure the efficacy of the behavioral intervention in the R00 phase of the project. Training in pragmatic trial design and management will

allow me to learn how to build interventions that are practical for constrained, real-world clinical settings, such as home hospice care. I will gain this expertise through coursework, training institutes, and engagement with my mentorship team.

Training Objective 4. Professional development: grant writing and collaboration. During my postdoctoral training to date, I have begun to develop my grant writing skills and collaborative network. I applied for and was awarded two grants for mentored pilot projects related to sociodemographic differences in EOL care for PwD. For both of these projects I have developed collaborative relationships across the New York Presbyterian hospital community and with community partners (e.g. Visiting Nurse Service of New York). However, I require further development in grant writing and collaborative networks to enable me to develop a sustainable line of research focused on improving care for dying PwD and FCG. During the K99 and R00 portions of this project I will participate in didactic workshops and national conferences. I will also work with my mentors to continue to hone my grant writing skills to submit a successful R00 application, with the ultimate goal of submitting an R01 application to the NIA. I will also work with my mentors and advisors and use professional channels to build a network of collaborators for future multi-site studies. I will complete Responsible Conduct of Research training at WCM and additional ethics training as outlined below.

Career Development Activities during the Award Period

The proposed research in this K99/R00 application builds on my prior training and expertise in disparities in EOL care and exposes me to new research topics, methods, and experiences to enhance my research skills and expertise. The training activities I propose for this award period will enable me to become a leading expert in developing practical, scalable and culturally inclusive solutions to improving EOL care and support for PwD and FCG. As a T32 Postdoctoral Fellow, I have learned about Geriatrics and clinical practice and developed grant writing skills. I am leading collaborative, locally-funded studies on disparities in EOL care for PwD in hospice and during terminal hospitalization. The content knowledge and research experience I have acquired as a T32 Postdoctoral Fellow have added to the theoretical and methodological expertise I gained as a doctoral student. To become an independent investigator in the area of improving care and support for PwD and FCG, I require additional knowledge and skills in several areas: knowledge of hospice, dementia caregiving, and recruitment and retention; behavioral intervention development and dissemination; and developing and running RCTs and pragmatic trials. Below, I outline the mentorship and training activities for each of 4 training objectives that are crucial to achieving my short-term goals of conducting the proposed research in this K99/R00 application and to my long-term goal of becoming a leading researcher in reducing disparities and improving care for PwD. This award will give me the time and support I need to develop as an independent investigator and allow me to collect pilot data, submit manuscripts, and ultimately, submit an R01 to reduce disparities and improve care for patients with dementia at the EOL.

Mentorship Team. I have assembled a strong, interdisciplinary team of mentors to provide guidance and support in my learning objectives, to provide and facilitate access to hands-on training, and to support the proposed research (Table 1 and Figure 1).

Holly Prigerson, PhD, my primary mentor, is Professor of Sociology in Medicine and Director of the Center for Research on End-of-Life Care at WCM. She has an extensive record of mentoring over 90 junior faculty into independent careers in academic medicine and has received awards for this mentorship. Dr. Prigerson is an internationally recognized EOL researcher with numerous NIH-funded studies addressing communication and **racial disparities in EOL care** for patients with cancer, including R01s. She served as PI on an NCI U54 inter-institutional partnership between Harvard and UMass Boston to address racial and ethnic disparities in cancer care. Dr. Prigerson has committed to continuing our mentorship relationship. As primary mentor on this grant, she will oversee all aspects of the study. In addition, she will provide focused mentorship in EOL care, **developing behavioral interventions**, and designing and **conducting RCTs** (Objectives 1-3). She will also provide mentorship in successfully running a research project as PI and writing and submitting a competitive R00 and R01 (Objective 4). She has agreed to provide additional resources and support staff (from her NIH R35 grant) as needed to support the K99 portion of my project including statistical, research and administrative support the details of

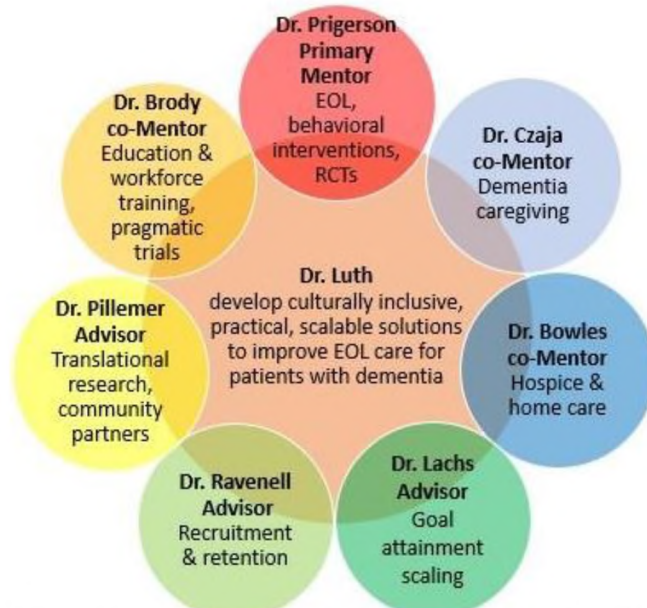


Figure 1. Complementary Expertise: mentors' and advisors' specific areas of expertise as related to Dr. Luth's long-term goal

which can be found in her letter. Dr. Prigerson will provide ongoing and sustained involvement throughout the project. She will continue our weekly hour-long meetings during the K99 portion of the project to provide **dedicated senior researcher attention** to all aspects of the study development, design, and analyses and to provide advice and support for career development. She will support me through submitting the R00 application and will meet with me quarterly during the R00 award to provide support.

Sara Czaja, PhD, a co-mentor, is Director of the Center on Aging and Behavioral Research at WCM. She is an internationally-recognized expert in **developing technology-based behavioral interventions for diverse populations of family caregivers of patients with dementia**. She has been continually funded by the NIH for over 30 years, including over 19 years of support for the multi-site Center for Research on Aging and Technology Enhancement (CREATE), which she recently brought to WCM. Dr. Czaja will provide focused mentorship in dementia caregiving, recruitment and retention of underrepresented minority groups, developing behavioral interventions, and designing and conducting RCTs (Objectives 1-3). She will also provide mentorship on grant writing and will facilitate connections with other dementia caregiving researchers (Objective 4). During the K99 award period, Dr. Czaja has committed to meeting with me on a monthly basis to provide input on study design, recruitment, analysis, and manuscript and grant preparation. She will also meet with me to respond to other issues that may arise if I think her input would be beneficial. Dr. Czaja will continue to meet with me quarterly for the duration of the R00 award to support the digitization of the tool.

Abraham Brody, PhD, RN, FAAN, a co-mentor, is Associate Professor of Nursing and Medicine and Associate Director of the Hartford Institute for Geriatric Nursing at New York University. He is a nationally recognized expert in developing efficacious, multi-modal, real-world interventions to **improve healthcare clinicians' abilities to care for persons with dementia in community-based settings** (Objectives 1 & 2). He has R01 and R61-R33 funding from the NIA to carry out **pragmatic trials** in 25 sites across the country to improve hospice care for patients with dementia (Objective 3). Dr. Brody agrees to meet with me at least quarterly during the K99 award, and more frequently as needed, to provide support on study design, recruitment, and manuscript and grant preparation and to discuss progress towards my professional development goals. He will involve me in training opportunities related to his work that deepens my understanding of workforce training and dementia (Objective 1) and will connect me with other researchers in our areas of interest (Objective 4). Dr. Brody will meet with me twice a year during the R00 award for ongoing support.

Kathryn Bowles, PhD, RN, FAAN, FACMI, a co-mentor, is Professor and vanAmeringen Chair in Nursing Excellence at the University of Pennsylvania School of Nursing and Vice President and Director of the Center for Home Care Policy & Research at Visiting Nurse Service of New York (VNSNY). She is an internationally-renowned expert in **improving outcomes for seriously ill older adults**, including developing behavioral interventions that target clinical practice and decision-making (Objectives 1 & 2). Dr. Bowles has committed to meeting with me at least quarterly for the K99/R00 award periods to discuss study design, advise on grant proposals, and **facilitate coordination within VNSNY** to ensure both phases of the research project can be completed.

Advisory Panel. I have gathered an expert advisory committee consisting of researchers with expertise that aligns with my study aims and training objectives. Dr. Karl Pillemer, PhD, is Professor of Gerontology in Medicine at WCM, and Hazel E. Reed Professor of Human Development and Senior Associate Dean for Research and Outreach in the College of Human Ecology at Cornell University. Dr. Pillemer has extensive experience in working with **community partners, translational research, workforce training, and caregiving** (Objective 1). He will also assist me with grant writing at the R00 and R01 stages (Objective 4). Dr. Joseph Ravenell, MD, is Associate Dean for Diversity Affairs at New York University. Dr. Ravenell will provide expertise in **recruitment and retention**, particularly with members of the **African American community** (Objective 1). Dr. Mark Lachs, MD, is the Irene F. and I. Roy Psaty Distinguished Professor of Clinical Medicine, Co-Director of the Center for Aging Research and Clinical Care, and Co-Chief of the Division of Geriatrics and Gerontology at WCM. Dr. Lachs is a nationally-recognized expert in elder care and elder abuse in a variety of domestic and institutional settings and will provide support on using **goal attainment scaling** to develop the tool for hospice clinicians (Objective 2).

Table 1. Competencies, Mentors and Advisors	Mentors				Advisors		
	Prigerson	Czaja	Brody	Bowles	Pillemer	Ravenell	Lachs
Competency							
End-of-life care	x			x			x
Hospice	x		x	x			
Dementia caregiving		x	x		x		
Recruitment and retention		x	x			x	
Intervention development	x	x	x	x			x
Workforce training			x	x	x		
Community partner collaboration					x		
Goal attainment scaling							x
RCTs and pragmatic trials	x	x	x	x	x		
Responsible conduct of research	x	x	x	x	x	x	x
Professional development	x		x		x		

of the **African American community** (Objective 1). Dr. Mark Lachs, MD, is the Irene F. and I. Roy Psaty Distinguished Professor of Clinical Medicine, Co-Director of the Center for Aging Research and Clinical Care, and Co-Chief of the Division of Geriatrics and Gerontology at WCM. Dr. Lachs is a nationally-recognized expert in elder care and elder abuse in a variety of domestic and institutional settings and will provide support on using **goal attainment scaling** to develop the tool for hospice clinicians (Objective 2).

My mentors and advisors will help me form a strong collaborative of leading researchers in EOL care, hospice, dementia caregiving, workforce training, recruitment and retention, behavioral intervention development, and RCT and pragmatic trial design so that I can achieve all of the learning objectives I outline above. Importantly, they also have established numerous collaborations, including co-mentorship, co-authorship and co-investigation on a variety of related projects among themselves, indicating how they work well together, though each will contribute to my research development in distinct ways. Mentorship will be most intense during the K99 phase of the award and will diminish during the R00 phase. Some interaction with mentors and advisors is maintained during the R00 portion of the award as light touchpoints to troubleshoot issues with study implementation (for example, Dr. Bowles facilitating collaboration with VNSNY) and to provide feedback on an eventual R01 submission. I will meet with all mentors and advisors by teleconference quarterly during the K99 phase and with all mentors semi-annually during the R00 phase.

Training Activities for K99. In addition to the didactic experiences and meetings described here, mentorship activities and competencies are outlined in the previous section and presented in Tables 1 and 2. Didactic experiences include various types of exposures to material including shadowing clinicians, online coursework, and in-person training and workshops. These experiences vary from half-day workshops to lengthier programs over the course of several days or months.

Training Objective 1. Extend knowledge in core substantive areas: hospice, dementia caregiving, and recruitment and retention (Mentors: Prigerson, Czaja, Brody, Bowles. Advisor: Ravenell). Didactic Experiences. In the first year of the K99 phase, I will shadow clinicians including geriatrics physicians and hospice nurses and social workers as they provide care to PwD to learn more about hospice care and dementia caregiving. In order to gain experience working with hospice organizations, I will become a trainer in Dr. Brody's R33 multi-site grant to provide education to hospice clinicians (letter: Brody). Meetings. In years 1-2 of the K99, I will attend WCM and VNSNY grand rounds and team meetings monthly to deepen my understanding of issues related to geriatrics, EOL, and hospice care; dementia caregiving; and how hospice organizations run. I will also attend weekly meetings at the Center for Research on End-of-Life Care to be actively involved in ongoing EOL care research projects at the center.

Training Objective 2. Develop, implement, and disseminate behavioral interventions (Mentors: Prigerson, Czaja, Brody, Bowles. Advisors: Pillemer, Lachs). Didactic Experiences. In the first year of the K99, I will apply to attend the NIH Training Institute on Dissemination and Implementation Research to learn about disseminating research findings to clinical practice and community partners. This is a 4-month online and 2-day in-person training on putting evidence-based strategies into clinical practice. If I am not accepted, I will attend the Cornell Translational Research Summer Institute, a 2.5-day training on translating research findings into practice. To learn about designing behavioral interventions, I will attend the Johns Hopkins Center for Innovative Care in Aging: Behavioral Interventions. This is an online or one-day face-to-face session on developing, evaluating, and implementing behavioral interventions. Meetings. In the second year of the K99 I will attend AcademyHealth: Science of Dissemination and Implementation in Health Conference. This is a 3-day conference designed to help researchers and practitioners bridge the gap between research, practice, and policy in health and health care.

Training Objective 3. Design and conduct clinical trials: RCT and pragmatic trials (Mentors: Prigerson, Czaja, Brody, Bowles). Didactic Experiences. In preparation for leading an RCT in the R00 phase and a pragmatic trial and in the R01 stemming from the K99/R00 award, in year 2 of the K99 I will apply to participate in the NIH Summer Institute on Randomized Behavioral Clinical Trials, a two-week program on planning, design, and execution of RCTs involving behavioral interventions. If I am not selected to attend this course, I will complete two courses at WCM: 1) Clinical Trials Design and Analysis, which provides a comprehensive overview of designing, conducting, and analyzing clinical trials; and 2) Clinical Trials: Trial Designs, Correlative Sciences, Ethical Considerations, Regulatory Oversight. Both are one-day workshops offered each Fall by the Clinical & Translational Science Center at WCM. I will also complete WCM's 4-week Protocol Writing Workshop, also offered each Fall. Finally, I will complete the NIH Pragmatic and Group-Randomized Trials in Public Health and Medicine, a 7-part online course on designing and analyzing pragmatic and group-randomized trials with videos, slide decks, suggested reading materials and guided activities.

Training Objective 4. Professional development: grant writing and collaboration (Mentors: Prigerson, Czaja, Brody, Bowles. Advisor: Pillemer). Didactic Experiences. During the second year of my K99 I will increase my focus on professional development and grant writing through various initiatives through WCM. I will retake WCM's Responsible Conduct of Research course, a semester-long, online and in-person (11 contact hours) course. I will also complete the WCM Planning and Writing Successful Grant Proposals, a day-long workshop

offered regularly to teach fundamentals of proposal writing and provide tips and strategies for writing grant proposals. I will apply for the WCM Leadership in Academic Medicine Program, a year-long series of monthly workshops to promote career planning, self-management, and leadership skills. Meetings. In years 1-2 of the K99, I will attend monthly work in progress sessions at the NIA-funded Roybal Center Translational Research Institute on Pain in Later Life. I will present my work to this group twice a year. I will also disseminate my research findings at national conferences throughout the duration of the K99/R00 award, including: Gerontological Society of American Scientific Meeting (years 1-5), American Association of Hospice and Palliative Medicine Annual Conference (years 3-5), and the American Sociological Association Annual Meeting (years 2 & 4).

Table 2. Training Objectives, Meetings, and Activities for K99 Award

Training Objective (Aims)	Mentorship and Advisor Meetings (Frequency)	Training Activities (Time)	Year/Semester ^a					
			Year 1			Year 2		
			1	2	3	1	2	3
1. Increase knowledge in core substantive areas (Aims 1-4)	• Dr. Prigerson (weekly) • Dr. Czaja (monthly) • Dr. Brody and Dr. Bowles (quarterly) • Dr. Ravenell (quarterly) (recruitment and retention)	<u>Didactic Experiences</u>						
		• Shadow geriatrics and palliative care physicians, hospice clinicians (10 days)	x	x	x			
		• Complete Hospice Champion Training with Dr. Brody (2 days)	x					
		<u>Meetings</u>						
		• Grand rounds and team meetings at WCM and VNSNY (monthly)	x	x	x	x	x	x
2. Develop, implement, and disseminate behavioral interventions (Aim 2)	• Dr. Prigerson (weekly) • Dr. Czaja (monthly) • Dr. Brody and Dr. Bowles (quarterly) • Dr. Pillemer (quarterly) (translational research, community partner collaboration) • Dr. Lachs (quarterly) (goal attainment scaling)	• Center for Research on End of Life Care research meetings (weekly)	x	x	x	x	x	x
		<u>Didactic Experiences</u>						
		• Apply to NIH Training Institute on Dissemination and Implementation Research (4 month online and 2 day in-person) OR Bronfenbrenner Center for Translational Research: Summer Institute (2.5 days)		x	x			
		• Johns Hopkins Center for Innovative Care in Aging: Behavioral Interventions (online or 1 day)				x		
		<u>Meetings</u>						
3. Design and conduct RCT and pragmatic trials (Aims 3-4)	• Dr. Prigerson (weekly) • Dr. Czaja (monthly) • Dr. Brody and Dr. Bowles (quarterly)	• Academy Health: Science of Dissemination and Implementation in Health conference (3 day)				x		
		<u>Didactic Experiences</u>						
		• WCM Protocol Writing Workshop (4 weeks)						x
		• Apply to OBSSR NIH Summer Institute on Randomized Behavioral Clinical Trials (2 weeks) OR WCM: Clinical Trials: Trial Designs, Correlative Sciences, Ethical Considerations and Regulatory Oversight and Clinical Trials Design and Analysis (1 day each)					x	x
		• NIH Pragmatic and Group-Randomized Trials in Public Health and Medicine (online)				x		
4. Professional development (Aims 1-4)	• Dr. Prigerson (weekly) • Dr. Czaja (monthly) • Dr. Brody and Dr. Bowles (quarterly) • Dr. Pillemer (quarterly) (career development, grant proposals)	<u>Didactic Experiences</u>						
		• WCM Responsible Conduct of Research (4 months)						x
		• WCM Planning & Writing Successful Grant Proposals (1 day)						x
		• Apply to WCM Leadership in Academic Medicine Program (1 year)				x	x	x
		<u>Meetings</u>						
		• Translational Research Institute on Pain in Later Life Works in Progress (monthly)	x	x	x	x	x	x
		• Gerontological Society of America, American Association of Hospice and Palliative Medicine, American Sociological Association conferences (annual)	x	x	x	x	x	x

a. Semester 1=Spring, 2=Summer, 3=Fall

Specific Aims

A growing number of older adults with Alzheimer's Disease and related dementias will die at home with hospice care.⁸⁻¹¹ Patients with primary or comorbid dementia comprise 50% of hospice recipients¹² and are more likely than patients with other diseases to experience suboptimal outcomes at end of life (EOL)¹³⁻¹⁵ (e.g. under-assessment and -treatment of symptoms,^{13,15} increased family caregiver (FCG) burden,¹⁶ higher rates of hospice disenrollment^{17,18}). African American patients with dementia (PwD) are at elevated risk for poor hospice outcomes, such as 2½ times greater risk of hospitalization. Most prior work on EOL care focuses on hospice care for cancer patients and white PwD in nursing homes.¹⁹⁻²² Providing home hospice clinicians with the training and tools they need to address dementia-related issues could help improve symptom management, reduce FCG burden, and decrease hospice disenrollment.²³⁻²⁷ Training in dementia care for home hospice clinicians and culturally inclusive, flexible tools to help them to identify and address the needs of PwD and FCG, particularly African American persons, are urgently needed as this population grows.

My overarching research interests are in developing practical, scalable solutions for improving care and support for home hospice PwD and their FCG while attending to and reducing sociodemographic disparities in outcomes. I seek NIA K99/R00 award funding to acquire the necessary training and skills to become an independent investigator, including: gaining deeper knowledge of hospice care for PwD; learning evidence-based techniques for intervention development, including learning from African Americans about how to incorporate their views into intervention development; and designing and conducting randomized controlled trials (RCTs) and pragmatic trials. The goals of this study are to 1) assess challenges in caregiving for PwD at EOL and 2) develop and pilot test a) a much-needed training program for home hospice clinicians to learn strategies to address common and distressing challenges for FCG of PwD and b) a novel goal-assessment tool capable of adapting to personalized needs of FCG. Following the NIA's Stage Model,²⁸ this study will provide the basic science (Stage 0), development (Stage 1A) and preliminary testing (Stage 1B) to determine the potential for both the training program and goal assessment tool.

Aim 1 (K99 Y1). To identify common challenges, strategies and gaps in care and support for community-dwelling PwD near the EOL. Sub Aim 1. Identify commonalities and differences in challenges, strategies, and gaps for African American and white FCG. Semi-structured interviews with African American (n=20) and white (n=20) FCG and hospice clinicians (n=20) will identify challenges, strategies, and gaps. A follow-up survey (n=80) of hospice clinicians will provide information on the frequency with which these occur at a large urban hospice provider.

Aim 2 (K99 Y2). To adapt culturally inclusive, dementia-focused training materials and create a goal assessment tool for home hospice clinicians to guide care and support for PwD and FCG. Sub Aim 2. Incorporate perspectives of key stakeholders from diverse racial backgrounds. The training will tailor existing materials on dementia caregiving to address the challenges and strategies specific to home hospice care for dying PwD identified in the literature and supported by Aim 1. The tool will encompass a culturally inclusive range of challenges, goals, and outcomes related to EOL dementia care as identified by diverse stakeholders that will guide clinicians in tailoring care to the challenges individual FCG face.

Aim 3 (R00 Y1-Y2). To examine feasibility and acceptability of the training and tool and revise them based on feedback. Hypotheses: >70% of home hospice clinicians will: 1) complete the training, 2) report the training and tool are useful, 3) increase knowledge of dementia-related challenges and care strategies, and 4) increase confidence in providing dementia-related care.

Aim 4 (R00 Y2-Y3). To conduct a pilot test to determine the preliminary efficacy of the training and tool. Hypotheses (compared to usual care): Clinicians who complete the training and use the tool (intervention group) will have greater knowledge of dementia-related challenges in EOL care and how to manage them effectively. Those in the intervention group will report it helps them identify individual PwD and FCG challenges and suggest appropriate management strategies. FCG of intervention group clinicians will report reduced FCG burden. PwD of intervention group clinicians will experience lower rates of hospice disenrollment. Exploratory analyses will examine African American-white disparities in FCG burden and PwD disenrollment.

The long-term goal of this research is to reduce FCG burden and improve care for PwD. The project develops and tests a culturally inclusive, clinically useful training and tool for home hospice clinicians to identify and suggest appropriate strategies for challenges FCG face providing care to PwD at the EOL. This training and tool will set the stage for a sustained line of research to enhance home hospice care for PwD and FCG, including a subsequent R01 application for a fully powered RCT to test more rigorously the efficacy of the training and tool to improve hospice care for PwD and to address disparities in care.

A. Significance

A.1 The number of patients with dementia (PwD) in home hospice is growing. The number of adults age 65+ in the United States with Alzheimer's Disease and related dementias (dementia) is expected to reach 13.8 million by 2050.²⁹ African American adults are twice as likely to be diagnosed with dementia as non-Hispanic whites.¹¹ Increasingly, PwD die at home and with hospice care.⁹ Up to 40% of PwD die at home⁸ and nearly half receive hospice care.⁹ Up to 50% of hospice patients have a primary or comorbid dementia diagnosis.¹² *There is an urgent need to provide high-quality home hospice care and support to the growing proportion of PwD and their family caregivers, particularly African American persons, who are at increased risk for dementia.*

A.2 PwD and their family caregivers (FCG) experience suboptimal EOL and hospice outcomes. While hospice can promote better quality of life for dying individuals,³⁰ it often fails to meet the unique needs of PwD and FCG. Over half of PwD experience pain at EOL,^{13,31} but are less likely than patients with no dementia to be prescribed palliative medication.¹⁴ African American patients are 60-70% as likely as whites to receive pain medication in the last week of life.¹⁵ PwD^{10,17,18} and persons from underrepresented racial groups¹⁸ are more likely undergo burdensome transitions out of hospice. In my pilot study, African American PwD were more likely than whites to be disenrolled from hospice. Family caregiving for PwD is also particularly burdensome. It requires more hours of care and additional responsibilities such as performing basic medical tasks, health care decision-making, and advocating for PwD.³²⁻³⁴ Although African American FCG of PwD report more positive appraisals of caregiving,^{35,36} caregiving may be particularly burdensome for African American FCG who experience discrimination in health care encounters.³⁷ *Aim 1 addresses the urgent need to understand why PwD, particularly African American persons, experience suboptimal outcomes at EOL and in home hospice.*

A.3 Community-dwelling PwD and FCG may have unique needs at the EOL. Both community-dwelling PwD and FCG have significant needs at early and middle stages of dementia.^{34,38-44} Cognitive impairment in PwD can make communication and pain assessment difficult.⁴⁵ Poor pain management can be exacerbated for African American patients who may also experience clinician underestimation of pain symptoms⁴⁶ or misconceptions that they have a higher pain tolerance.⁴⁷ Lack of illness understanding, more common among African Americans,⁴⁸ is linked to requesting and receiving non-beneficial EOL care.^{49,50} Not understanding dementia progression is a barrier to receiving effective EOL care in PwD⁵¹ and may cause FCG to request non-beneficial care resulting in hospitalization. Neuropsychiatric symptoms of dementia, present in half of hospice-eligible PwD,⁵² contribute to PwD hospitalization and FCG stress.^{23,53-58} Although prior research rigorously addresses the needs of community-dwelling PwD and FCG at earlier stages of dementia⁵⁹⁻⁶³ and EOL for PwD in nursing homes,^{22,64,65} there is a gap in knowledge about the needs of community-dwelling PwD and FCG near the EOL, particularly those from underrepresented groups. *Aim 1 will identify the most common and distressing challenges faced by African American and white FCG of community-dwelling PwD near EOL.*

A.4 Goal attainment scaling is a feasible approach to addressing the lack of interventions to support racially diverse home hospice PwD and their FCG. There is robust prior research that outcomes for PwD (e.g. behavioral disturbances⁵⁴) and their FCG (e.g. perceived burden^{54,59}) can be improved. However, few successful interventions have been developed for or tested in home hospice settings⁶⁶ and none focus on the challenges African American families face. Home hospice nurses, social workers, and chaplains (clinicians) are well-positioned to help identify common and distressing challenges faced by PwD and FCG and offer strategies to address these challenges.²⁴ However, they lack specific training and tools to do so.²³⁻²⁷ Goal attainment scaling (GAS) offers a feasible, reliable, valid, and responsive, and easy-to-use outcome measure for frail PwD.⁶⁷⁻⁷¹ GAS tools are flexible and can be completed by FCG, and/or clinicians.⁷⁰ The content of goals and outcomes can be tailored to the individual vary.⁷⁰ As such, GAS tools can be individually tailored to respond to the heterogeneous circumstances related to EOL dementia caregiving. Although it has not been tested in hospice settings or with underrepresented racial groups, GAS's appropriateness for frail PwD and its flexibility give it the potential to respond to differing needs of persons from diverse backgrounds and with different beliefs. *Aims 2-4 propose to develop, refine, and pilot test a training and GAS tool for hospice clinicians to identify and recommend strategies for common and distressing challenges for PwD and FCG.*

This project addresses the need for clinically relevant, culturally inclusive training and tool for home hospice clinicians to detect and address common and distressing issues for the growing number of PwD and FCG.

B. Innovation

The proposed project is innovative in the following ways. First, while there is a substantial body of research about the experience of PwD in nursing homes^{19-21,64} this project **will fill a gap in knowledge about an understudied, but growing population: community-dwelling home hospice PwD and FCG** (Aim 1). Second, the project **identifies and will address the needs of African American individuals**, whose experiences are underrepresented in existing research (Aims 1-4). Third, this is the first project to **apply goal attainment scaling to a new setting** (hospice care) and incorporates **new populations** (underrepresented

racial groups). Goal attainment scaling has been successfully applied in other clinical settings,^{67,69-71} but has not yet been tested in hospice care or for its cross-cultural applicability. Finally, the project develops and tests a **new clinically relevant, user-friendly, culturally inclusive training and tool** that supports hospice clinicians in identifying and addressing the individual needs of PwD and FCG (Aims 2-4).

C. Approach

C.1 Overview. This study has two overall goals. First, it will identify and quantify the challenges FCG and home hospice providers face when caring for PwD. Second, it will develop and pilot test training materials and a goal assessment tool for home hospice clinicians. The training and tool will support clinicians to address common and distressing dementia-related caregiving challenges for African American and white FCG of PwD. As explained in Sections C3-C6, this study follows Stages 0, 1A and 1B of the NIA Stage Model of Behavioral Intervention Development.²⁸

This study will use a mixed-methods design including qualitative interviews, a quantitative survey, and intervention development and testing (Figure 2). To achieve **Aim 1** (K99) we will conduct interviews with African American and white FCG and hospice and palliative care clinicians of PwD nearing EOL. We will ask about the most common and distressing challenges of dementia caregiving at the EOL and strategies to address those challenges. We will conduct a follow-up survey of home hospice clinicians to estimate the frequency with which challenges, gaps, and strategies exist in an urban hospice provider. For **Aim 2** (K99) we will obtain multiple rounds of structured, key stakeholder input from FCG, hospice clinicians, and research/content experts. We will use this input to develop culturally inclusive training materials and a goal assessment tool for clinicians. The training and tool will help home hospice clinicians identify and suggest tailored strategies to address common and distressing challenges for PwD and FCG. For **Aim 3** (R00), we will test the training and tool with African American and white FCG and hospice clinicians to determine feasibility and acceptability. We will revise the training and tool based on feedback. For **Aim 4** (R00) we will conduct a pilot study to determine the preliminary efficacy of the training and tool in home hospice clinicians and FCG of their PwD. Efficacy outcomes include: improving clinician knowledge and confidence (primary outcome), reducing FCG burden, and decreasing PwD hospice disenrollment (secondary outcomes). We will compare African American-white differences in outcomes for FCG and PwD.

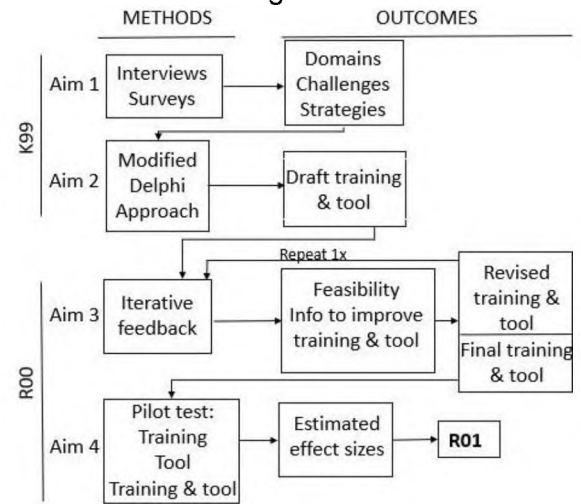


Figure 2. Research Approach

C.2 Pilot Data and Team Expertise. My experience in EOL care, racial and ethnic disparities, and dementia places me in a strong position to carry out this work. I have also assembled a robust and accomplished team of mentors and collaborators with highly relevant, interdisciplinary expertise. My primary mentor, Dr. Prigerson, is an established researcher in the areas of EOL care, racial and ethnic disparities, and intervention development and implementation. I have assembled a team of co-mentors and advisors with multidisciplinary expertise in the areas I have identified for development. These areas include: knowledge of hospice care (Prigerson, Brody, Bowles), dementia caregiving (Czaja, Brody), and recruitment and retention in African American communities (Czaja, Brody, Ravenell); behavioral intervention development and dissemination (Prigerson, Czaja), particularly in the areas of workforce training (Brody, Bowles, Pillemer), translational research and community partner collaboration (Pillemer), and goal attainment scaling (Lachs); and developing and running RCTs and pragmatic trials (Prigerson, Czaja, Brody, Bowles) (see Figure 1 and Table 1).

C.3 Aim 1 (K99 Y1). Identify challenges, strategies, and gaps in care and support for PwD and FCG.

Procedure. An expansive literature documents how PwD die in nursing home settings and how to support community-dwelling PwD and FCG in early and middle stages of dementia (e.g. CASCADE^{22,64,65} and REACH⁵⁹⁻⁶³ studies). Little research has focused on the care and support needs of community-dwelling PwD and FCG nearing the EOL and their FCG. To fill this gap and consistent with Stage 0 of the NIA's model,²⁸ we will conduct semi-structured interviews with African American and white FCG and healthcare providers who provide care to community-dwelling PwD near the EOL. Data collection of these interviews is already underway due to an internal WCM grant on which I am the PI. Interviews will last 30-45 minutes. We will ask participants to identify common and distressing challenges when caring for PwD, strategies they use to address challenges, and gaps in care and support. We will ask about issues that are common in dementia caregiving (e.g. understanding dementia progression, communicating with PwD, FCG burden) and allow participants to identify additional challenges. We will ask participants to share their strategies for addressing challenges and about their experiences with strategies prior research identifies as helpful. Sampling equal numbers of African American and white FCG will allow us to compare similarities and differences in

challenges, strategies, and perceived gaps in dementia. We will then conduct an online survey of 80 clinicians at a hospice provider to estimate how often challenges and strategies occur at that hospice.

Sample and Recruitment. Interviewing 40 FCG (20 African American, 20 white) and 20 healthcare providers (geriatrics and hospice physicians, nurses, social workers, chaplains, etc.) will provide us with sufficient data to identify the most common and distressing challenges and strategies in dementia caregiving. We will also identify similarities and differences between African American and white FCG. If we do not reach thematic saturation with 40 FCG, we will continue interviewing FCG until we do. We will recruit FCG from two Weill Cornell Medicine-NewYork Presbyterian (WCM-NYP) outpatient settings that provide care to 1,500 community-dwelling PwD from diverse backgrounds each year. We will also recruit FCG with the help of CaringKind, our community partner with 30 years of experience engaging with thousands of African American and white FCG of PwD. We will screen FCG to ensure they are caring for PwD nearing the EOL. We will recruit provider interview participants from WCM-NYP divisions of Geriatrics and Palliative Medicine and General Internal Medicine and Visiting Nurse Service of New York (VNSNY) home hospice. For the survey, we will recruit hospice clinicians from VNSNY home hospice. A target of 80 survey respondents is expected based on previous experience in this population. See letters: WCM-NYP, CaringKind, VNSNY.

Outcomes. We expect FCG and providers will identify dementia-related issues they encounter providing care to community-dwelling PwD nearing the EOL. We expect African American and white FCG will identify both overlapping and different issues. Aim 1 will give us the domains and initial content to guide material selection for the training and content development for the tool.

C.4. Aim 2 (K99 Y2). Adapt training materials and develop a tool for home hospice clinicians.

Procedure. Hospice clinicians are important in ensuring PwD and FCG receive caring and supportive attention at EOL. However, hospice clinicians rarely receive training in how to address the unique situations and needs that arise in dementia caregiving at EOL.²³⁻²⁷ Moreover, hospice clinicians do not have tools to support patients and FCG with dementia-specific needs.²³⁻²⁵ To fill this gap and consistent with Stage 1A of the NIA model,²⁸ we will develop a clinician training to enhance knowledge and practical tool to support their clinical practice to address PwD's and FCG's dementia-related needs at EOL.

Training content will be based on literature^{34,38-40} and the challenges and strategies identified in Aim 1. Table 3 provides a draft schema for the training, including dementia-related domains, topics within each domain, and example strategies. Cultural inclusivity will be a theme that is incorporated and repeated across all sections of the training. To limit participant burden, we will provide training sessions as in-service modules. Training will be modeled on a training program that increased healthcare workers' knowledge, improved communication, and altered clinical practice.⁷²⁻⁷⁴ We will use evidence-based techniques for increasing healthcare workers' knowledge and changing practice behaviors, including: repetitive exposure to information, case studies, and interactive activities.⁷⁵

Table 3. Sample Training Components

Cultural Inclusivity		
Domains	Topics	Strategies
Implicit Bias	Recognizing and addressing implicit bias	Stereotype replacement, taking perspective
Dementia at EOL	What happens as dementia progresses Arranging/Managing care for PwD at EOL	Talking points for FCG conversations Items to ask about for treatment goals
Communicating with PwD	<u>End-stage</u> : developing a care plan for FCG <u>Less advanced</u> : finding out what PwD wants	Questions for understanding PwD preferences Tips for interacting with PwD
FCG stressors for dementia at EOL	Anticipatory grief, emotional ambivalence about impending death, PwD & FCG safety Resources for FCG of PwD	Referrals, information about resources & support services for FCG of PwD (<i>customized to location of care</i>)

We will base the tool on goal-attainment scaling (GAS), a feasible, reliable, valid, and responsive outcome measure for frail older PwD.⁶⁷⁻⁷⁰ GAS can respond with flexibility to the heterogeneous nature of dementia caregiving at EOL including stage/severity of dementia, primary/comorbid diagnoses, availability/nature of FCG support, PwD ability to communicate, disease progression, and sociocultural contexts. FCG may experience some or all of these, with varying degrees of distress. GAS's flexibility gives it the potential to respond to differing needs of persons from diverse backgrounds and with different beliefs. Clinicians will help FCG to identify goals to address the dementia care-related issues that are most distressing for them and then set expected and plausible outcomes that are better or worse. The goals will complement and can be incorporated into standard hospice care planning. The tool will be structured around domains of dementia caregiving at EOL identified in the literature and in Aim 1. Table 4 provides sample goals and outcomes for the domain "FCG stressors." To limit burden on clinicians and FCG, we will develop a pre-determined menu of goals with expected, better, and worse outcomes.⁷⁰ If they prefer, FCG will be able to create goals and outcomes.

We will seek feedback from FCG and clinicians and collaborate with CaringKind to develop the training and tool content for hospice clinicians (see letter). To develop the training modules and tool content, we will use a

Table 4. Sample goals and outcomes

Domain: FCG Stressors with dementia at EOL				
Goal: FCG uses support system to make time for self.				
-2	-1	0	+1	+2
Worst outcome	Worse outcome	Expected outcome	Better outcome	Best outcome
Action 1: Make referral to respite care. Place initial call on behalf of FCG.				
Respite care not initiated	Respite care 1-2x/ month	Respite care weekly	Respite care 3x/ week	Respite care daily
Action 2: Church contacted for support				
No church support	Church support 1-2x/ month	Church support weekly	Church support 3x/ week	Church support daily

strategies. Participants will be asked to return the feedback within 4 weeks. We will review and compile responses and return to key informants for additional feedback, which we will gather in a 30-60 minute follow-up telephone, video-conferenced, or in person interview within the following 2 weeks.

Sample Size and Recruitment. We will recruit 15-20 key stakeholders (target: 5 hospice clinicians, 5 African American FCG, 5 white FCG, 5 content/research experts) from the interview and survey samples. Members of the research team and collaborators will serve as content/research experts. If interview and survey participants are not want to assist with training and tool development, we will engage with hospice and community partners to assist with recruitment. Based on our previous experience with this approach,⁷⁸ 20 participants are enough to ensure sufficiently comprehensive feedback from multiple perspectives. See letters: CaringKind, VNSNY.

Outcomes. This aim will provide us with two products to examine feasibility and acceptability in Aim 3: 1) Dementia-enhanced training for home hospice clinicians; 2) GAS-based tool for hospice clinicians and FCG to set goals and outcomes for dementia-related care and support needs at the EOL.

C.5. Aim 3 (R00 Y1). Examine feasibility and acceptability of the training and tool and finalize them.

Procedure. We will use a two-stage, iterative approach to determine the feasibility and acceptability of the training and tool and obtain feedback for improving them (NIA Model Stage 1B²⁸). During each of the two stages of testing, we will present the training or tool to clinicians and FCG in interactive sessions. We will present the training to clinicians and ask for feedback as we progress through the training. We will then conduct a debriefing exercise to clarify feedback received during the training. We will obtain participants' perceptions on feasibility of the training including utility of the individual modules and training appropriateness and fit within organizational culture. We will evaluate the tool using a "think aloud" strategy with FCG and hospice clinicians.⁷⁹ After each think aloud exercise, participants will complete a 15-20 minute qualitative interview to confirm and clarify findings from the exercise.⁷⁹ Participants will assess the tool's feasibility, including cultural inclusiveness, usability, and perceived utility. At the end of each stage, we will modify the training and tool based on results from the interactive sessions, resulting in a revised (after stage 1) and final (after stage 2) training and tool (see Figure 2). We will create manuals for each and digitize the final tool using CREATE protocols for user-centered design and usability.⁸⁰

Sample Size and Recruitment. We will test feasibility and acceptability and revise the training and tool with a total of 50 clinicians and FCG. We will pilot test the training with two groups of 10 hospice clinicians (n=20). We will pilot test the tool with two groups of 15 (5 hospice clinicians, 5 African American FCG, 5 white FCG) (total n=30). Given that 10-30 participants are adequate for this type of study^{81,82} sample sizes of 20 and 30 are expected to be sufficient to ensure sufficiently comprehensive feedback from multiple perspectives. See letter: VNSNY confirming support in recruiting clinicians and FCG.

Outcomes. Aim 3 will result in a final training and digitized tool for preliminary efficacy testing in Aim 4. The training and tool will be culturally inclusive and practical for incorporation into hospice care delivery.

C.6 Aim 4 (R00 Y2-Y3). Conduct pilot test to determine the preliminary efficacy of training and tool.

Procedure. We will compare the training and tool (intervention) and usual care (control) to determine the intervention's preliminary efficacy to 1) improve hospice clinicians' knowledge about dementia caregiving at EOL, 2) reduce FCG burden, and 3) reduce PwD hospice disenrollment. We will conduct the pilot test with VNSNY hospice clinicians, who are organized into geographic teams of 8-12 persons. We will combine VNSNY's clinical teams into groups of approximately 20 clinicians. We will randomly assign one group of clinicians to the intervention group and one to the control group (n=40, 20 in each group). We will deliver the intervention to clinicians as in-service sessions during team meetings. Clinicians in the control group will receive an in-service presentation on patterns of hospice use among PwD. We will assess clinicians' knowledge about dementia caregiving at EOL before and after the in-service sessions. We will work with clinicians and VNSNY research staff to identify and enroll 4 African American and white PwD with a FCG for each clinician in the study (n=160, 80 in each group). We will focus on FCG of PwD who have been in home

modified Delphi approach,⁷⁶ soliciting multiple rounds of structured feedback from a team of key stakeholders. Stakeholders include hospice clinicians, African American and white FCG, and content/research experts). We will circulate feedback among stakeholders until consensus is reached.⁷⁷ Participants will be asked to comment on the clarity and relevance of training and tool questions and the feasibility of suggested

hospice for more than 1 week, as these PwD are more likely to live for the duration of the study (4 weeks). Clinicians in the intervention group will use the tool with FCG to set goals related to dementia care at baseline. Clinicians will track progress towards those goals at up to three additional visits with the FCG (visits 2-4). We will assess FCG burden prior to each of these 4 visits. Clinicians in the control group will not have access to the training materials or tool and will deliver usual care. We will assess FCG burden among FCG of clinicians in the control group up to 4 times, each time prior to a clinician visit. We will ask FCG of all clinicians in the study to report PwD hospice discharge status 3 and 6 months after study enrollment. We will verify FCG reports using electronic hospice records. A randomized controlled trial (RCT) is considered a rigorous design for pilot testing preliminary efficacy of interventions.⁸³ We will register this pilot RCT on ClinicalTrials.gov so that other researchers may reproduce and extend our findings.

Sample Size and Recruitment. We will test the preliminary efficacy of the training and tool with 40 hospice clinicians at VNSNY. We will group VNSNY's hospice teams into groups of relatively equal size (target n=20). We will randomly assign each group into one of two conditions: intervention or control. To maximize participation, we will deliver the training as in-service sessions during hospice team meetings. We will assess baseline and follow-up measures for 160 FCG (80 African American, 80 white). See letter: VNSNY.

Outcomes. Aim 4 will give pilot data, including directionality and effect size estimates for a fully-powered RCT.

C.7 Measures. Table 5 contains a list of proposed measures, schedule, and participants. All measures in Table 5 will be assessed via self-report. We will also verify hospice discharge status using electronic hospice records. Wherever possible we will use validated or objective measures.^{69,84-91} However, for some outcomes, no validated measures exist. In these cases, we will draw upon measures used in the published literature,^{50,92-94} develop measures based on published recommendations, or develop measures tailored to our project and that may be validated in future studies. We will assess additional measures as well. **Aim 1.** Types of interview questions are described in section C3 and Appendix 1. Appendix 2 contains draft survey questions and format. Final content will be based on interviews. **Aim 3.** We will assess the most and least helpful aspects of the training and tool. Success of training will be assessed by 70% or greater endorsement of each feasibility measure. **Aim 4.** We will assess usability of tool in real-world hospice settings using Dr. Czaja's CREATE usability protocols.⁸⁰

Domain	Proposed measure (items)	Aims	Schedule	Participants
Outcomes				
Feasibility	Utility, cultural inclusivity, usability (TBD)	3	Post	C, FCG
Knowledge	Dementia knowledge assessment scale (27) ⁸⁴ Communicating with PwD strategies (TBD)	3,4	P/P,B, V2,V3,V4	C, FCG
FCG burden	Zarit Caregiver Burden Inventory (12) ^{85,86}	1-4	B,V2,V3,V4	FCG
Discharge status	FCG report of PwD status, verified in hospice record	4	3M, 6M	FCG
Covariates				
Demographic Information	Age, gender, race, ethnicity, marital status, employment status, education, income, religion, insurance coverage	1-4	B	C, FCG
	Brief RCOPE (14) ⁸⁷	1-4	B	FCG
Experience with healthcare	Perceived discrimination scale (17) ⁸⁸	1-4	B	FCG
	Perceived discrimination in healthcare encounters (1) ⁹²			
Attitudes	Black-White IAT; ^{89,90} Attitudes towards blacks scale (20) ⁹⁴	3,4	Pre	C
Knowledge	Terminal illness acceptance (1) ^{50,93}	4	B,V2,V3,V4	FCG
Goal attainment	GAS Score ^{69,91}	4	B,V2,V3,V4	FCG

TBD=to be developed. P/P=pre/post. B=baseline. V=visit. M=month. C=clinician. FCG=Family caregiver.

C.8 Data Analysis. Qualitative analysis. For qualitative methods, we believe sample sizes will be sufficient to obtain thematic saturation. If necessary, we will continue to collect data at all stages until thematic saturation is achieved. Data coding and analysis will be informed by a responsive interviewing model in which blocks of information are combined based on the theme they represent.⁹⁵ Trained raters will independently review the transcripts and identify passages that include suggestions for inclusion and/or modifications to the intervention. Raters will then organize these passages into categories reflecting single themes, discuss these themes, and make revisions until consensus is reached regarding themes. Identified themes will inform modifications to the intervention. Additionally: **Aim 1.** We will record and transcribe interviews and code transcripts for themes. **Aim 2.** Two members of study team will be present to take notes during stakeholder meetings. Notes and written feedback from stakeholders will be incorporated into training and tool development. **Aim 3.** Two members of study team will be present to take notes during training presentation and "think aloud" exercises. Focus groups following training and interviews following "think aloud" sessions will be recorded, transcribed and coded for themes. Data will be incorporated into revising and finalizing the training and tool.

Quantitative Analysis. We will calculate descriptive statistics to characterize the sample of participants in each aim. **Aims 3-4.** We will calculate changes in participant outcomes by comparing individual responses to questions at baseline and various points in time after exposure to the training and tool. **Aim 4.** To establish preliminary efficacy, we use logistic regression analysis. The outcome variables are described in section C7 and include dichotomous measures of improvement in clinician knowledge, reduction in FCG stress, and reduction in hospice disenrollment. The main predictor will be participation in the intervention vs. control

groups. Our sample of 160 FCG will allow us to detect an effect size with an odds ratio of 2.33 with a power ($1-\beta$) of 0.82 at $\alpha=0.05$. With a sample size of 40 for clinicians we will not have sufficient power to detect a small effect size. However, the purpose of the pilot testing is to provide directionality and estimates for effect sizes so that a fully-powered RCT can be conducted in future research (R01 submission).

C.9 Alternative Approaches. A. We do not include PwD in interviews because but the reliability of the data and need to create sensitive, reliable tools for assessing them is beyond the time, resources, and scope of this 1-year investigation. We will explore including PwD in future studies. B. The limited budget and short time frame of the study precludes development of a web-based or other training format that may be equally or more effective. We can test alternate approaches in future studies. C. We could use a different approach to develop the tool than goal attainment scaling, which has not been used previously in hospice care or tested with racially diverse populations. However, goal attainment scaling has been successfully used with older adults and PwD and has the flexibility to be appropriate for different racial groups.

C.10 Challenges and Response. A. Recruiting 10 FCG from the 40 interview participants may be a challenge. Response: By gauging FCG interest in helping with training and tool development as we conduct interviews we can identify difficulties with recruitment early in the project. We will engage with hospice and community partners to assist with recruitment (letters: VNSNY, CaringKind). B. We may have difficulty enrolling African American FCG, who are typically underrepresented in research. Response: We will work with WCM-NYP physicians (letters: Pelzman, Adelman) and community partners (letter: CaringKind) who serve African American PwD. We will also seek input from experts in recruiting in African American communities (letters: Brody, Ravenell) and prioritize African American recruitment over the recruitment of white subjects. C. Challenges and strategies identified in interviews could be an artifact of individuals who are interviewed. Response: The survey will allow us to characterize frequency of issues, strategies, gaps beyond interviewees. D. Utilization of goal assessment tool may be imbalanced among African American and white PwD. Response: We will include clinicians who have both African American and white patients on their caseloads in pilot testing and help clinicians identify which patients on caseloads to use when testing the tool.

C12. Research

Timeline. Table 6 outlines the timeline for research activities across the award period.

C.11 Benchmarks for

Success. A. We will obtain 4 publications in peer-reviewed journals:
1) study of caregiver and clinician perspectives on challenges, strategies, and gaps in home

Table 6. Research Timeline (Quarters)

Aim	Activity	K99								R00											
		Year 1				Year 2				Year 3				Year 4				Year 5			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Startup	IRB approval, RA training	x	x																		
Aim 1: Interviews, survey	Participant recruitment		x	x	x																
	Data collection and analysis			x	x	x															
Aim 2: Develop training, tool	Develop training and tool with stakeholders					x	x	x	x												
Aim 3: Feasibility testing	IRB approval									x											
	Participant recruitment									x	x	x	x	x							
	Feasibility test and refine training and tool										x	x	x	x	x						
Aim 4: Preliminary efficacy testing	IRB approval													x							
	Participant recruitment													x	x	x	x	x			
	Data collection and analysis														x	x	x	x	x		
Ongoing	Manuscript preparation					x				x				x	x	x	x	x	x	x	
	R00/R01 applications					x	x	x	x									x	x	x	x

hospice care for PwD (K99), 2) developing training and tool (K99), 3) pilot testing and revising training and tool (R00), and 4) preliminary efficacy of training and tool (R00). B. We will present study results annually at national conferences and through invitations to speak locally to relevant stakeholder groups and institutions. C. We will have sufficient data and information for a competitive R01 application for a fully powered RCT to test training and tool in other hospice settings (rural) and/or with larger or different populations (e.g. more hospices, Latinx FCG). D. I (Dr. Luth) will launch as independent investigator with necessary expertise and skills to become a leading expert in improving hospice care for PwD and FCG.

C.13 Rigor of Prior Research and Current Approach. Rigorous research exists on supporting FCG and community-dwelling PwD at early and middle stages of dementia⁵⁹⁻⁶³ and the EOL for PwD in nursing homes.^{22,64,65} However, very few projects focus on PwD in home hospice care, and none focus on the needs of African American PwD and FCG. This project addresses the lack of research on PwD in home hospice, specifically African American persons. We will use an RCT, a methodologically rigorous approach,⁸³ to determining the preliminary efficacy of the intervention we propose to develop and pilot test. We will register this pilot test on ClinicalTrials.gov so that other researchers may reproduce and extend our findings.

C.14 Sex as a Biological Variable and Authentication of Resources. Participants will not be excluded based on sex or gender. Recruitment rates of PwD, FCG and clinicians across sex/gender will be monitored during the study to minimize selection biases. Given that over 60% of FCG and 72% of hospice clinicians are female,^{36,96} we expect the majority of our FCG and clinician participants to be female. In the event that a patient sex/gender is under-represented (<30%), we will modify recruitment strategies to target the under-represented group. This study does not use biological or chemical resources.